LunaAura: An Explainable Machine Learning Framework for Hormonal and Emotional Health Analysis

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***Abstract*—Hormonal rhythms, daily behaviour, and psychological stress together shape emotional wellbeing, yet most digital wellness tools still treat them as separate streams of information. Menstrual and mood-tracking applications typically record symptoms and visualise past trends, but rarely offer predictive insight into how lifestyle choices interact with the menstrual cycle to influence mood. This work describes *LunaAura*, an explainable machine learning framework that models day-level emotional wellbeing from menstrual-cycle indicators, self-reported mood and stress, and sleep–lifestyle variables. Publicly available datasets on cycles, sleep habits, and mental health are merged into a cycle-aware, longitudinal dataset with engineered features capturing hormone–stress and sleep–stress interactions. A Random Forest classifier is trained to estimate an emotional-risk label, and SHAP (SHapley Additive Explanations) is used to quantify global and instance-level feature contributions. The analyses indicate that mood score, perceived stress, luteal-phase indicators, and interaction features are the strongest drivers of predicted risk, while sleep duration and a sleep–stress ratio behave as protective factors. The results illustrate how explainable, cycle-aware models can support transparent emotional-health analytics.**

***Index Terms*—Digital phenotyping, emotional wellbeing, explainable AI, menstrual cycle, Random Forests, SHAP values**

I. Introduction

Emotional wellbeing rarely depends on a single factor. For menstruating individuals, cyclical hormonal changes, daily stressors, sleep habits, and broader lifestyle patterns all contribute to how a given day feels. Clinical work has shown that symptoms of mood and anxiety disorders often intensify in specific menstrual phases, particularly in the days leading up to menstruation, and that this effect can be clinically significant even when baseline symptoms are modest [1]. At the same time, poor sleep and chronic stress are well-established risk factors for depression and anxiety in the general population.

Most widely used digital tools only partially reflect this complexity. Period-tracking apps provide useful records of cycle dates and symptoms, but their analytics are largely descriptive. Mood-tracking apps collect emotional ratings and sometimes offer generic feedback based on moving averages. Few systems attempt to learn how a person's stress, sleep, and menstrual phase combine to affect their emotional state on a particular day.

Digital phenotyping research has shown that machine learning models trained on smartphone and wearable data can detect stress, anxiety, and mild depression with promising accuracy [2], [3]. However, the menstrual cycle is rarely treated as the central temporal structure around which symptoms fluctuate. The LunaAura project addresses this gap by building an explainable, cycle-aware classifier for day-level emotional-risk prediction.

II. Related Work

A. Menstrual-Cycle Analytics and Mental Health

Large-scale analyses of period-tracking apps have demonstrated that self-tracked data can be used to characterise menstrual health and fertility patterns at population scale. Symul *et al.* analysed millions of daily observations and showed that app-based charting can robustly estimate ovulation timing and highlight deviations from typical patterns [4]. These studies concentrate on reproductive outcomes rather than emotional health but illustrate the power of longitudinal self-tracking.

The relationship between menstrual phase and mental health has long been recognised. The American Psychiatric Association summarises evidence that a substantial subset of patients experience premenstrual exacerbation of mood or psychotic symptoms and recommends cycle-informed assessment as part of routine psychiatric evaluation [1]. Ross *et al.* used intensive daily monitoring to show that suicidal ideation and planning can peak in the perimenstrual window for some women with a history of suicidality [5]. Ferguson *et al.* argue that encouraging patients to track cycles alongside mood can support suicide prevention [6].

B. Cycle-Aware Prediction and Explainable AI

AlSaad *et al.* developed models to predict mood swings in reproductive-age women using metabolic, menstrual, and lifestyle indicators collected via structured questionnaires; ensemble methods outperformed linear models and SHAP highlighted symptom burden and lifestyle factors as leading predictors [7]. Choi *et al.* review digital phenotyping approaches for stress, anxiety, and depression using smartphone and wearable data [2], [3].

Explainable AI has become central in healthcare. Lundberg and Lee introduced SHAP as a unified, game-theoretic framework for interpreting model predictions [8]. Mienye *et al.* and Mardani *et al.* survey XAI in healthcare and clinical decision

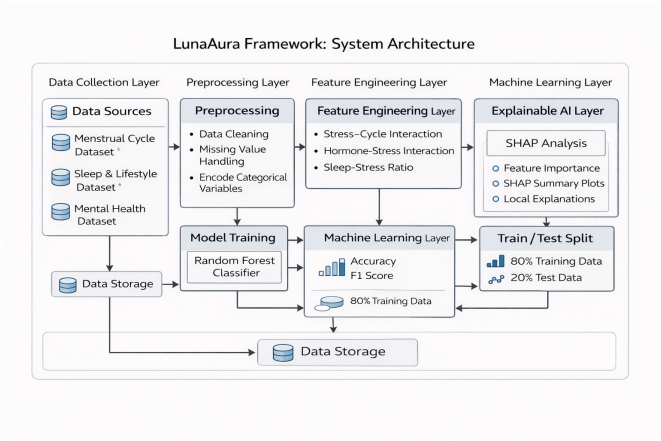

Fig. 1. System architecture of the LunaAura framework.

support, emphasising transparency and workflow integration [9], [10]. Heinrichs *et al.* focus on practical challenges when bringing XAI into clinical practice and stress that explanations must be concise, context-aware, and trustworthy [11].

Within this landscape, LunaAura combines a tabular Random Forest classifier with SHAP explanations, with a specific focus on menstrual-cycle context and lifestyle factors.

III. System Architecture

LunaAura follows a layered machine learning architecture with data collection, preprocessing, feature engineering, model training, and explanation layers.

The data-collection layer aggregates three sources: a menstrual-cycle dataset, a sleep and lifestyle dataset, and a depression or mental-health dataset. The preprocessing layer standardises formats, aligns dates, handles missing values, and encodes categorical variables. The feature-engineering layer constructs domain-informed interaction features that combine cycle position, stress, and sleep. The machine-learning layer trains the Random Forest classifier and evaluates its performance using standard metrics. The explainable AI layer applies SHAP analysis and produces visualisations such as global feature-importance plots and summary plots [9], [10].

IV. Dataset Description

A. Integrated Day-Level Dataset

The three source datasets (menstrual-cycle, sleep/lifestyle, and depression or mental-health) are merged into a single day-level integrated dataset. Each row corresponds to one calendar day for a given individual, with columns drawn from all three sources where available. Date fields are used to align records; only days with sufficient information across the sources are retained.

Cycle-related fields are derived from the menstrual dataset: **cycle_day** gives the day index within the current cycle, **cycle_phase** assigns each day to a broad phase (menstrual, follicular, ovulatory, luteal), and **luteal_flag** marks late-luteal days, which are often clinically sensitive periods [1]. Sleep-related fields, such as **sleep_duration**, originate from the sleep/lifestyle dataset. Emotional-health fields, including a daily **mood_score** and depression-related indicators, originate from the mental-health dataset.

TABLE I
Representative Features in the Integrated Dataset

Feature	Description
mood_score	Self-reported emotional wellbeing
stress_level	Daily perceived stress (ordinal)
cycle_day	Day index within menstrual cycle
cycle_phase	Menstrual / Follicular / Ovulatory / Luteal
luteal_flag	Late-luteal indicator (binary)
sleep_duration	Night-time sleep hours
physical_activity	Coarse activity level proxy
hormone_proxy	Cycle-position-based hormone proxy

TABLE II
High-Level Statistics of the Integrated Dataset

Statistic	Description
Total observations	Number of merged day-level entries
Cycle length mean	Average menstrual cycle duration (days)
Sleep duration mean	Average sleep hours per night
Stress distribution	Frequency of low/medium/high stress
Risk day proportion	Percentage of emotional-risk labels

A binary emotional-risk label is created at the day level. Days with low mood and/or depression-related flags are labelled as risk days, and all other days form the non-risk class. Because risk days tend to be less common, the label distribution is imbalanced.

B. Variables and Features

Table I lists key features used in the model.

C. Dataset Statistics

Descriptive statistics for the merged dataset are summarised in Table II. Exact numerical values depend on the specific public datasets used, but the structure remains the same. The total number of day-level records depends on how many days have aligned entries from all three sources. Average cycle length provides context on whether the sample reflects typical menstrual-cycle durations. Average nightly sleep duration and the distribution of stress categories provide a high-level view of lifestyle factors. The proportion of emotional-risk days indicates the degree of class imbalance.

Because emotional-risk days are less frequent than non-risk days, the dataset exhibits class imbalance. A classifier that simply predicts the majority (non-risk) class would achieve deceptively high accuracy while failing to identify many risk days. To address this, the Random Forest uses class-balanced weights so that misclassifying minority-class days is penalised more strongly during training.

V. Feature Engineering

A. Cycle-Aware and Lifestyle Features

Cycle-aware features include **cycle_day**, one-hot encoded **cycle_phase** indicators, and **luteal_flag**. These variables allow

the model to distinguish, for example, a high-stress day in the mid-follicular phase from a similar-stress day in the late-luteal phase. Sleep and lifestyle features such as **sleep_duration** and **physical_activity** capture elements of daily behaviour that can modulate emotional state.

The **hormone_proxy** feature is derived from cycle position (for example, normalised cycle day or encoded phase labels). It is designed as a coarse indicator of typical hormonal context, rather than a detailed physiological model. No direct hormone measurements are available, and the proxy should be interpreted as a position-based approximation rather than a sinusoidal or biochemically calibrated curve.

B. Interaction and Non-Linear Features

Several engineered features capture interactions and non-linearities:

$$\text{stress_cycle_interaction} = \text{stress_level} \times \text{cycle_day}, \tag{1}$$

$$\text{hormone_stress_interaction} = \text{hormone_proxy} \times \text{stress_level}, \tag{2}$$

$$\text{sleep_stress_ratio} = \frac{\text{sleep_duration}}{\text{stress_level} + 1}, \tag{3}$$

$$\text{stress_squared} = \text{stress_level}^2. \tag{4}$$

These are implemented as additional columns in the tabular dataset. The stress–cycle term reflects the idea that the same reported stress may have different consequences depending on cycle position. The hormone–stress term encodes the hypothesis that hormonal context modulates stress sensitivity. The sleep–stress ratio gives higher values when sleep is relatively high for a given stress level, reflecting a potential buffering effect of sleep. The squared stress term allows the model to capture non-linear escalation when stress is very high.

VI. Mathematical Formulation

Let $x_i \in R^M$ denote the feature vector for day i and $y_i \in \{0, 1\}$ the corresponding emotional-risk label, where $y_i = 1$ indicates a risk day. The aim is to learn a function

$$f : R^M \to [0, 1] \tag{5}$$

such that $f(x_i)$ estimates the probability that $y_i = 1$.

With binary cross-entropy loss ℓ, empirical risk minimisation can be written as

$$\min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \ell\big(y_i, f(x_i)\big), \tag{6}$$

where $\mathcal{F}$ is the hypothesis space of Random Forest models. A Random Forest with T trees represents f as

$$f(x) = \frac{1}{T} \sum_{t=1}^{T} h_t(x), \tag{7}$$

where $h_t(x)$ is the probability predicted by the t-th decision tree. Class labels are obtained by thresholding $f(x)$, typically at 0.5 for analysis, though this threshold can be adjusted to trade off recall and precision.

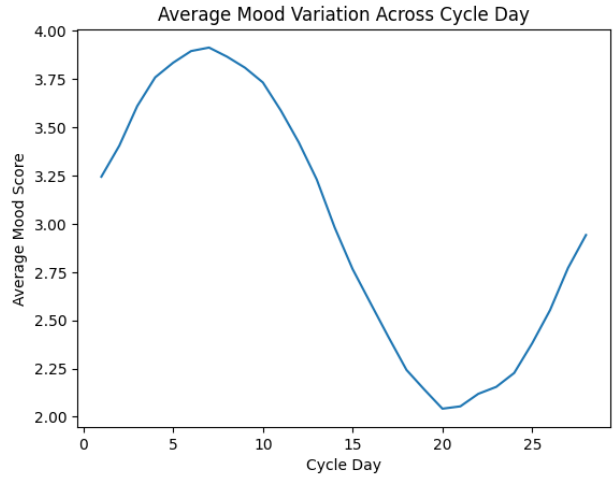

Fig. 2. Average mood variation across cycle days.

VII. Experimental Setup

The integrated day-level dataset is randomly partitioned into training (64%), validation (16%), and test (20%) subsets. Where user identifiers are available, all days for a given individual are kept within a single split to avoid information leakage from personal baselines. Stratified sampling on the emotional-risk label ensures that each split reflects the overall class imbalance.

All experiments are implemented in Python using `pandas` for data manipulation, `scikit-learn` for the Random Forest, and the `shap` library for explainability. Hyperparameters such as the number of trees, maximum tree depth, and minimum samples per leaf are tuned via grid search using five-fold cross validation on the training set. Class-balanced weights are enabled in the Random Forest classifier to mitigate label imbalance.

The main evaluation metrics on the held-out test set are accuracy, precision, recall, F_1 score, and ROC AUC, together with a confusion matrix and ROC curve. These metrics provide complementary perspectives on performance for the majority and minority classes.

A. Computational Complexity

The computational cost of training a Random Forest with T trees on n samples can be approximated as

$$\mathcal{O}\big(T \times n \log n\big), \tag{8}$$

because each decision tree performs a series of split searches that involve sorting subsets of the data [8]. For the moderate-sized tabular dataset used here, this complexity is acceptable: training completes in a short time on a standard workstation, and inference consists of traversing a limited number of trees. This makes the approach suitable both for offline research and for potential integration into interactive tools.

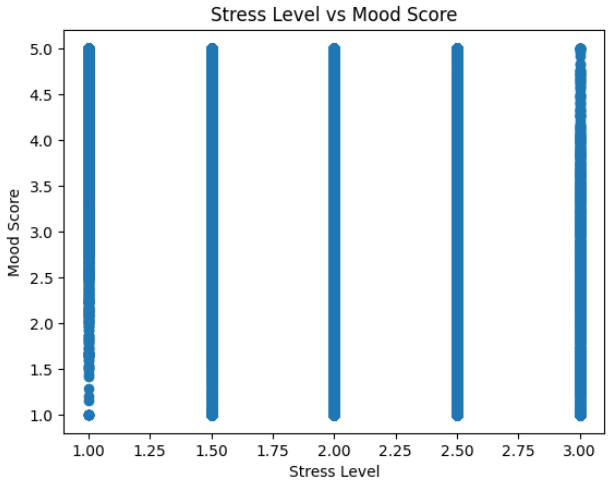

Fig. 3. Relationship between perceived stress level and mood score.

VIII. Exploratory Data Analysis

Figure 2 shows average `mood_score` across cycle days. Higher average mood is observed in the mid-follicular period, with a dip in the late-luteal days, which is consistent with reports of premenstrual mood deterioration [1]. This pattern supports the decision to include explicit phase information and the `luteal_flag` feature.

The scatter plot in Fig. 3 suggests a negative association between `stress_level` and `mood_score`. However, the vertical spread at each stress level indicates that other factors, such as cycle phase and sleep, modulate this relationship. Some high-stress days still have relatively good mood, while some low-stress days show poor mood, hinting at moderating influences not captured by stress alone.

IX. Model Evaluation

A. Global Performance and ROC Curve

Figure 4 shows Gini-based feature importance scores. The highest importance is assigned to `mood_score`, followed by cycle-related variables and stress-related features. This ranking is consistent with prior mood-swing prediction studies that also identify symptom severity and lifestyle variables as key predictors [7].

The confusion matrix in Fig. 5 summarises correct classifications and errors. True negatives correspond to correctly recognised non-risk days, and true positives correspond to correctly recognised risk days. False negatives represent risk days the model failed to flag, which is concerning in sensitive mental-health applications; false positives represent additional alerts on relatively lower-risk days. In a real deployment, the decision threshold could be tuned to reduce false negatives at the cost of more false positives, depending on user and clinician preferences.

The ROC curve in Fig. 6 illustrates the trade-off between true positive rate and false positive rate across different thresholds. The curve lies above the diagonal baseline for a wide range of thresholds, indicating discriminative ability noticeably better than chance. An ROC AUC in the low 0.8 range suggests that the model separates risk and non-risk days reasonably well, even though labels are noisy and based on self-report. For applications where missing risk days is particularly costly, operating points towards the upper-left portion of the ROC curve would be preferred.

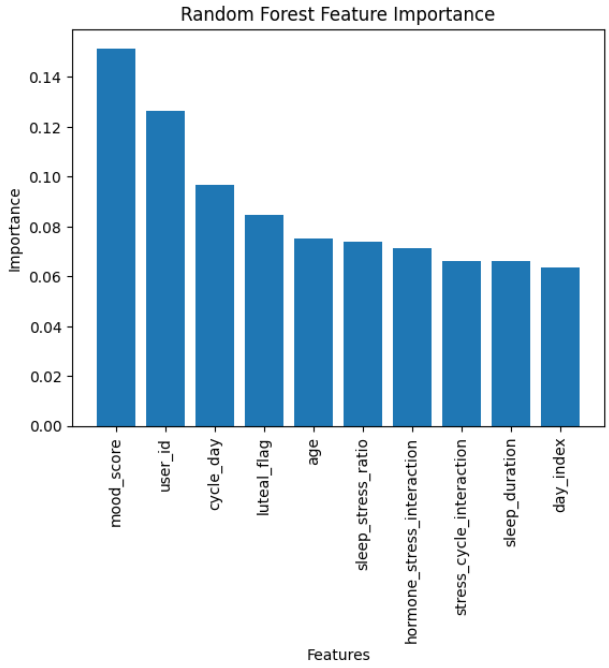

Fig. 4. Random Forest feature importance based on Gini importance.

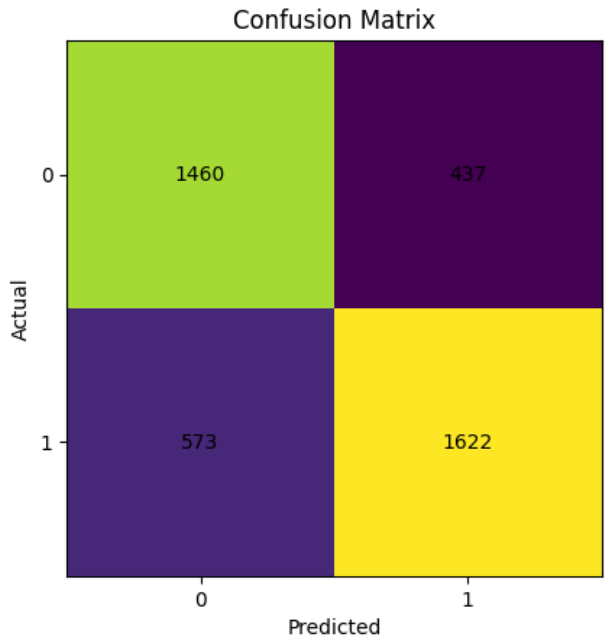

Fig. 5. Confusion matrix for the Random Forest classifier.

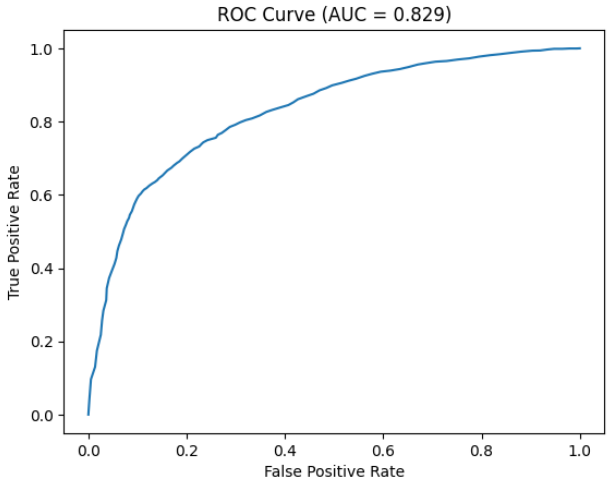

Fig. 6. ROC curve for the Random Forest classifier.

TABLE III
CONCEPTUAL BASELINE MODELS FOR LUNAAURA

Model	Characteristics	Limitations for this dataset
Logistic Regression	Linear classifier with interpretable coefficients	Cannot capture nonlinear hormone–stress and sleep–stress interactions without extensive manual feature engineering
SVM	Kernel-based margin classifier	Sensitive to feature scaling and hyperparameters; less transparent for non-technical users
Random Forest	Ensemble of decision trees	More complex but naturally models nonlinear interactions and mixed feature types

B. Baseline Models (Conceptual Comparison)

Although the Random Forest is the primary model in LunaAura, it is helpful to compare it conceptually with simpler baselines. Table III contrasts logistic regression, Support Vector Machines (SVMs), and Random Forests in terms of characteristics and limitations for this dataset.

Logistic regression is attractive for its simplicity but ill-suited to capturing the non-linear effects and interactions expected in hormonal and emotional health data. SVMs with non-linear kernels can handle some of this complexity but are harder to interpret and require careful tuning. Random Forests handle categorical and continuous variables with limited preprocessing, capture nonlinear effects through tree structure, and can be combined with SHAP for rich explanations, which makes them a reasonable choice in this context.

TABLE IV
CONCEPTUAL ABLATION OF FEATURE GROUPS

Feature Configuration	Expected Effect
Full feature set	Best predictive performance; captures cycle, stress, and sleep effects jointly
Without cycle-phase features	Reduced ability to capture luteal-phase risk and other phase-specific patterns
Without sleep variables	Reduced modelling of sleep as a buffer against stress-related risk
Without interaction features	Reduced ability to capture nonlinear hormone–stress and sleep–stress relationships

X. ABLATION STUDY ON CYCLE-AWARE FEATURES

To better understand the contribution of different feature groups, the project includes a conceptual ablation study. This analysis is conceptual in the sense that it discusses the expected impact of removing certain features, rather than reporting results from running additional experiments on reduced feature sets.

As Table IV indicates, removing `cycle_phase` and `luteal_flag` would weaken the model's capacity to detect phase-specific risk patterns observed in the exploratory plots and in clinical literature [1]. Omitting sleep variables such as `sleep_duration` and `sleep_stress_ratio` would limit the model's ability to distinguish high-stress days that are partly buffered by good sleep from those that are not. Dropping interaction features would simplify the hypothesis space and likely reduce performance on complex patterns, such as days when elevated stress during vulnerable phases has a disproportionate impact. These considerations highlight that cycle-aware and interaction features are not optional embellishments but central components of the modelling strategy.

XI. SHAP INTERPRETATION

Figure 7 shows global feature importance based on mean absolute SHAP values. The ranking broadly agrees with Gini importance but includes the advantage that SHAP accounts for feature interactions and correlations [8]. The highest SHAP scores are associated with `mood_score`, `luteal_flag` and related cycle-phase indicators, and stress-related features and interactions. Sleep-related variables and the sleep–stress ratio also contribute, though to a lesser degree.

The SHAP summary plot in Fig. 8 adds directional information by colouring points according to feature values. For `mood_score`, low values are associated with positive SHAP contributions (increasing predicted risk), while high values contribute negatively (reducing risk). For `luteal_flag`, the presence of the flag (late-luteal days) tends to push predictions towards the risk class, particularly when combined with high stress. For `sleep_stress_ratio`, higher values usually produce negative SHAP values, suggesting that adequate sleep

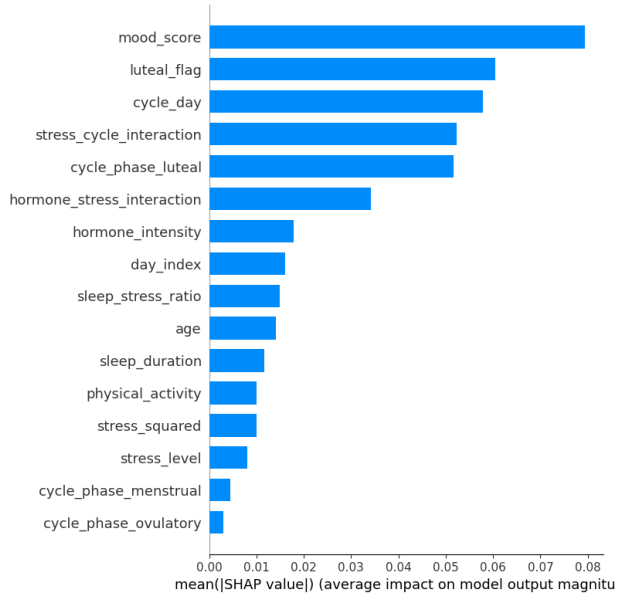

Fig. 7. Global feature importance based on mean absolute SHAP values.

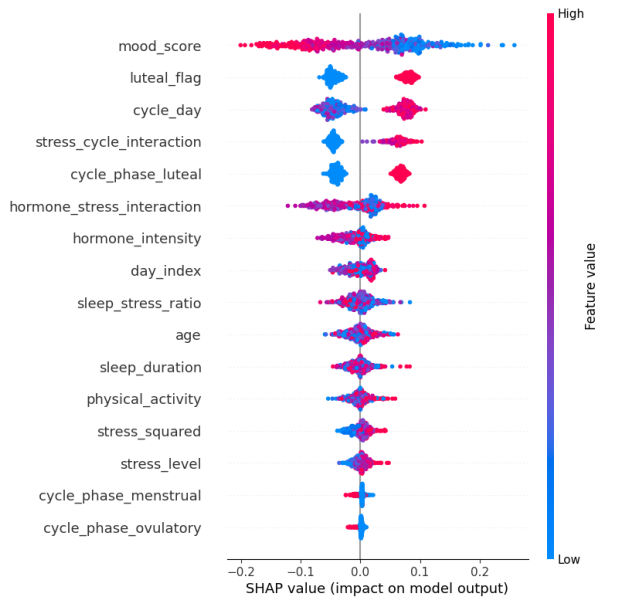

Fig. 8. SHAP summary plot showing feature-wise impact and direction.

relative to stress has a protective effect. These patterns align with the intended design of the engineered features and with clinical intuition [1], [7].

Local SHAP explanations for individual days (not shown in the figures) provide case-level insight into why a specific prediction was made. For a flagged high-risk day, the explanation typically highlights a combination of low mood, high stress, late-luteal status, and short sleep, whereas for a low-risk day with some stress, protective factors such as good sleep and non-luteal phase play a greater role. Such explanations can support human judgement by making the model's reasoning more transparent [11].

XII. Discussion

The LunaAura framework illustrates how cycle position, lifestyle factors, and emotional self-reports can be integrated into a single day-level dataset and analysed using an explainable Random Forest classifier. The presence of a clear late-luteal dip in average mood, combined with SHAP evidence that luteal indicators increase predicted risk, reinforces existing clinical observations that some individuals are especially vulnerable in this phase [1], [5]. The importance of stress-level interactions and the sleep–stress ratio suggests that the impact of stress is not uniform but depends on both cycle context and the presence of restorative factors such as sleep.

The results also underline the limitations of the current approach. Labels are derived from self-reported mood and depression-related indicators, which are inherently noisy and capture only part of the emotional experience. The model is trained on a moderate-sized, merged dataset and may not generalise perfectly to other populations, especially if demographic or clinical characteristics differ. Nonetheless, the combination of sensible performance, interpretable feature contributions, and alignment with existing literature indicates that the modelling choices are reasonable for an exploratory student research project.

XIII. Applications and Limitations

Potential applications of LunaAura-style models include personal feedback tools for individuals who already track their cycles, decision support for clinicians working with patients who report cyclical mood patterns, and research platforms exploring menstrual-related emotional dynamics. In each case, model outputs would be one input among many rather than an autonomous decision-maker.

Several limitations should be noted. The datasets are merged from different sources and may not represent a single clinically characterised cohort. Emotional state is compressed into a binary risk label, which simplifies analysis but loses nuance. Raw sensor data (such as continuous heart-rate or movement data) are not included, even though such signals might carry additional predictive information [2], [3]. The model may be biased if certain demographic or clinical groups are under-represented.

Algorithmic bias and privacy are important ethical concerns. Menstrual and emotional-health data are highly sensitive, and combining them with wearable data could enable intrusive inferences if governance is weak. Any real-world deployment should follow data-minimisation principles, use strong encryption, and give users fine-grained control over data collection and sharing [9]–[11].

XIV. Future Work and Conclusion

Further work could extend LunaAura in several directions. From a data perspective, prospective longitudinal studies with richer symptom diaries, hormonal assays, and wearable sensor streams would allow a closer examination of hormone–mood

relationships. From a modelling perspective, sequence models that explicitly track multi-day dynamics could be compared with the current day-level Random Forest while retaining SHAP or related explanation techniques.

Another natural direction is to integrate cycle-aware models into just-in-time adaptive interventions that adjust feedback and coping strategies based on predicted risk and menstrual phase [12]. Any such extensions would require careful clinical and ethical evaluation, particularly around how and when to surface risk information to users.

The LunaAura project shows that it is feasible to construct an explainable, cycle-aware classifier for emotional risk using publicly available datasets, basic feature engineering, and a Random Forest with SHAP explanations. While there is substantial room for refinement, the approach demonstrates a practical route towards interpretable menstrual-cycle-aware analytics in student-level machine learning research.

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